

Progress report

Project scope update

Since starting the project, I decided to narrow down my data to the state level. I also chose more specific features I want to analyze and explore: *days during the past 30 days where mental health was not good*, *violent crime*, *poverty rates per state*, *percentage with no insurance per state*, and *unemployment rate percentage per state*. My project still aims to look at the general patterns between mental health and crime rates, but it will also analyze the impact of socioeconomic variables. I will conduct a general EDA, correlational analysis, and clustering analysis to better understand the relationship. This will help us infer general conclusions about mental health and crime.

Data Sources

First, I am using a public API URL from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) Mental Health Indicators. The JSON link, which is included inside the code in config.py file, is [https://data.cdc.gov/resource/5eh7-pjx8.json?\\$limit=50000](https://data.cdc.gov/resource/5eh7-pjx8.json?$limit=50000). This API does not require a key. This data started with 1086 rows, but after filtering for states and questions, there were 54 rows. The key factor I am analyzing is the % value for the question: Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?

Second, I am using data from the Bureau of Justice Statistics' Crime Known to Law Enforcement 2024. I used GitHub Gists to upload the CSV into a Web CSV with a URL, which is stored inside the code in the config.py file. Otherwise, the data can also be found here: https://gist.githubusercontent.com/lrli1/d560e4d04b6b6e8315900bbe3224fc0c/raw/9dce7b56a93cc5e03e763b87e2dd5e56053525e0/Crime_Data.csv. This data started with 53 rows, but since there were missing states, it ended with 45. The key feature I am analyzing is the rate per 100,000 estimate, which represents the number of violent crimes (rape, robbery, aggravated assault, homicide) standardized per 100,000 people in the population

Lastly, I am using census data from the American Census Survey, 5-year estimates. I am looking at a table titled 'DP03 | Selected Economic Characteristics' for all states in the years 2021-2025. I used GitHub Gist to upload the CSV into a web environment at the following link: <https://gist.githubusercontent.com/lrli1/820584669eb70989d4d8d20b47d65e52/raw/809ccd245292102bcbae9a8c2c51f537462376f8/gistfile1.txt>. This link is also already stored inside the code. This data started with 53 rows, but ended with 52 due to extra headers. The features analyzed in this dataset include % of families whose income in the past 12 months is below the poverty level, % of the population with no health insurance coverage, and % of unemployment status for populations 16 and over in the labor force

Issues/Difficulties

In my crime data, four states were missing: Alaska, Arizona, Hawaii, Mississippi, and Pennsylvania. Since my analysis requires merging all tables together, every result has only 45 states instead of 50, which could skew results. Additionally, many socioeconomic variables that significantly influence both violent crime and mental health were not included. This can be seen by the similarities between states clustered together in the second clustering analysis. States in each group share similar patterns in urbanization, population density, political party, and more. Other factors that could be important include race, gender, education, etc. Because mental health and crime are so multi-dimensional, it is hard to represent the full story and implement policy without conducting a causal inference. My analysis is limited to understanding correlation and associations, which may not be tangible enough for actionable results.